LESSON PLAN

HIGH SPEED FLIGHT (12)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Effect of propeller rotation on aircraft and methods used to counteract it
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Effect of propeller rotation on aircraft and methods used to counteract it
5. Specific Objectives	12.1.1.1 Trainee will be able to describe the various types of propellers, principles of operation and pitch change mechanisms.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of P-Yaw and P-Roll diagrams; Instructor supervision on theoretical application only
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a propeller?
(b) Define P-Yaw and P-Roll in the context of propeller rotation.
(c) Which aircraft systems are typically used to counteract P-Yaw?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principles of P-Yaw generation • Principles of P-Roll generation • Aerodynamic causes of asymmetry • Impact on aircraft stability • Role of slipstream asymmetry	L & Ds	PPT, IB	15 min
(b)	• Function of P-Yaw countermeasure • Function of P-Roll countermeasure • Trimming procedures for P-Yaw • Trimming procedures for P-Roll • Effect of engine power changes	L & Ds	PPT, PDF	20 min
(c)	• Operational limits for countermeasures • Maintenance checks for trim systems • Troubleshooting P-Yaw issues • Troubleshooting P-Roll issues • Safety considerations during trim	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how P-Yaw affects the aircraft's directional stability.
(b) Describe the mechanism used to counteract P-Roll in flight.

12. Home Assignment and Follow Up	Prepare a brief note on the constructional details of the Dornier aircraft propeller, focusing on blade construction and pitch change mechanisms, to be covered in LP 05.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Constructional details about Dornier A/C propeller, construction of blades and pitch change mechanism and spinner	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min